

Describing data

Centre, spread, and why every ML pipeline starts here.

THE IDEA

Before you can model data you have to describe it. Two questions cover most of it: **where is the centre**, and **how spread out is it?**

Measure	What it tells you
Mean	The average. Add everything, divide by how many. Pulled by extreme values
Median	The middle value once sorted. Ignores extremes — better for skewed data like income
Variance	The average <i>squared</i> distance from the mean. Big variance = widely scattered data
Standard deviation	The square root of the variance — back in the original units

WHY VARIANCE SQUARES THE DISTANCES

If you just averaged the raw distances from the mean, the positives and negatives would cancel and you would always get zero. Squaring removes the signs. The square root at the end undoes the squaring so the answer is readable again — if your data is in years, the standard deviation is in years, but the variance is in years².

WORKED EXAMPLE

TOPIC 1 — DESCRIBING DATA

centre, spread, and why ML cares

OUR DATA — five patient ages

2, 4, 4, 6, 9

MEAN (the average)

add them up, divide by how many

$$2 + 4 + 4 + 6 + 9 = 25$$

$$25 / 5 = 5$$

MEDIAN (the middle value)

sort them, take the middle one : 2 4 [4] 6 9

$$\text{median} = 4$$

median ignores extreme values — mean does not

VARIANCE (how spread out)

step 1 : distance of each value from the mean (5)

$$2-5 = -3 \quad 4-5 = -1 \quad 4-5 = -1$$

$$6-5 = 1 \quad 9-5 = 4$$

step 2 : square each one (removes the minus signs)

$$9 + 1 + 1 + 1 + 16 = 28$$

step 3 : divide by how many

$$\text{variance} = 28 / 5 = 5.6$$

STANDARD DEVIATION

just the square root of the variance :

$$\sqrt{5.6} \approx 2.37$$

back in the original units — that is why we use it

ML LINK : normalising data is $(x - \text{mean}) / \text{std}$

it makes features with different scales comparable,
and it is the first step of almost every ML pipeline.

Why it matters for ML: normalisation is $(x - \text{mean}) / \text{std}$. Every feature you feed a model gets this treatment so that a feature measured in thousands does not drown one measured in decimals. Mean and standard deviation are also what "batch normalisation" normalises.

KEY POINTS FROM THIS TOPIC

Mean	add all values, divide by how many
Median	the middle value once sorted
Variance	average of the squared distances from the mean
Standard deviation	$\sqrt{\text{variance}}$ — back in original units
Normalisation	$(x - \text{mean}) / \text{std}$

TEST — Topic 1

Dataset: **3, 5, 7, 9, 11**

1. Find the mean and the median. → **mean = 7, median = 7**
2. Find the variance. → **distances -4,-2,0,2,4 → squares 16,4,0,4,16 = 40 → 40/5 = 8**
3. Find the standard deviation, then normalise the value 9. → **std = $\sqrt{8} \approx 2.83$; $(9 - 7)/2.83 \approx 0.71$**